UNIT 5 · 9 hours

Inverting and Non-Inverting Amplifiers



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

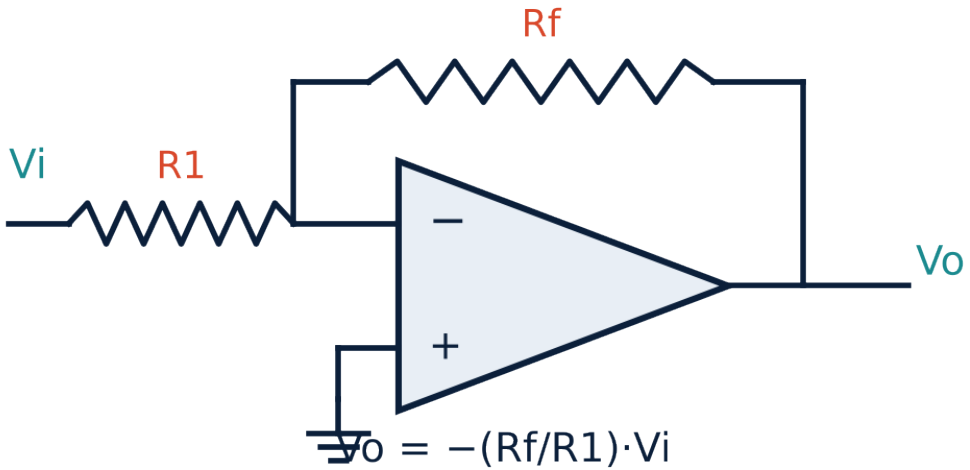
- Derive the gain of the inverting amplifier
- Derive the gain of the non-inverting amplifier
- Compare input resistance and phase of both
- Design amplifiers for a required gain

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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Inverting amplifier



- Input via R1 to the inverting terminal; + terminal grounded.
- Rf feeds output back to the – input.
- Virtual ground at the – terminal.
- Output is 180° out of phase.

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Deriving the inverting gain

1

Input current

i = (vi − 0)/R1 (virtual ground at 0 V).

2

No input current

No current enters the op-amp, so the same i flows through Rf.

3

Feedback

i = (0 − vo)/Rf.

4

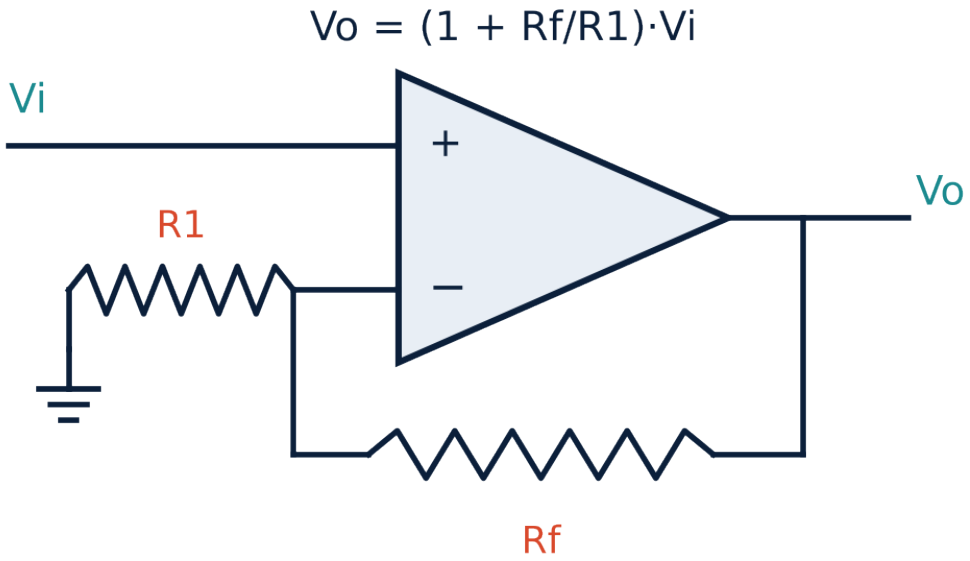
Equate

vi/R1 = −vo/Rf ⇒ vo/vi = −Rf/R1.

ACL = −Rf / R1, input resistance Rin = R1

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Non-inverting amplifier



- Input applied to the + terminal.
- R1 from − terminal to ground; Rf from output to −.
- The voltage at − equals vi (virtual short).
- i = vi/R1 = (vo − vi)/Rf ⇒ vo/vi = 1 + Rf/R1.
- Output in phase; Rin very high.

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Inverting versus non-inverting

Inverting

- Gain = −Rf/R1 (can be < 1)
- Phase shift 180°
- Input resistance = R1
- Virtual ground at the input

Non-inverting

- Gain = 1 + Rf/R1 (≥ 1)
- No phase shift
- Input resistance very high
- Voltage follower: Rf = 0, R1 = ∞ → gain 1

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Numerical 1: inverting amplifier

Problem R1 = 20 kΩ, Rf = 100 kΩ, vi = 1 V. Find vo, Rin and the input current.

- vo = −(Rf/R1) vi = −(100/20)(1) = −5 V
- Rin = R1 = 20 kΩ
- iin = vi/Rin = 1 V/20 kΩ

Answer: vo = −5 V, Rin = 20 kΩ, iin = 0.05 mA

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Numerical 2: design

Problem Design (a) an inverting amplifier with Rin = 30 kΩ and gain −10; (b) a non-inverting amplifier of gain 10 using R1 = 1 kΩ.

- (a) R1 = 30 kΩ; Rf = 10 × 30 kΩ = 300 kΩ
- (b) 1 + Rf/R1 = 10 ⇒ Rf = 9 R1 = 9 kΩ
- Check (b) with vi = 0.5 V: vo = 10 × 0.5 = 5 V

Answer: (a) Rf = 300 kΩ; (b) Rf = 9 kΩ

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Key takeaways

- Inverting: vo = −(Rf/R1) vi; Rin = R1
- Non-inverting: vo = (1 + Rf/R1) vi; high Rin
- Virtual short gives both results with simple algebra
- Voltage follower: unity gain buffer

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Practice questions

Q1. Derive the gain of an inverting amplifier.

Q2. R1 = 10 kΩ, Rf = 47 kΩ: find the gain of both configurations.

Q3. Design a non-inverting amplifier of gain 5.

Q4. What is a voltage follower and why is it used?